

SCALABLE VECTOR GRAPHICS

Animation In Action

<https://code.h5jun.com/fizo>



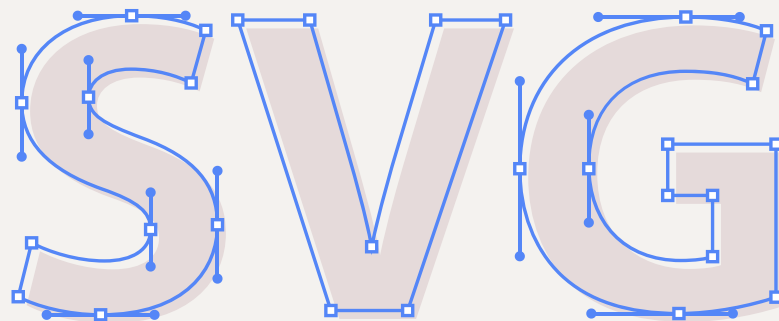
About me

李喆明 / @lizheming

- Github: <https://github.com/lizheming>
- Blog: <https://zh.eming.li>

SVG 是一种用来描述二维矢量图形的 **XML** 标记语言。因为都是W3C标准，SVG与其他的W3C标准，比如 **CSS**, **DOM** 和 **SMIL** 等能够协同工作。

<https://developer.mozilla.org/zh-CN/docs/Web/SVG>

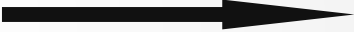


SVG 的基础

动画

变换 or 变形

动画

变换 or 变形 过渡  动画

SVG Animation

- SMIL
- JavaScript
- CSS Animation
- Web Animation API

SVG Animation

- SMIL 
- JavaScript
- CSS Animation
- Web Animation API

变换动画(Transform)

1. 平移 translate
2. 旋转 rotate
3. 缩放 scale
4. 斜歪 skew
5. 矩阵变换 matrix

平移变换

JS BinSave

HTMLCSSJavaScriptConsoleOutputHelp

HTML ▼

```

<!--mirror from https://codepen.io/...
<meta name="description" content="...
<div class="container">
  <div class="logo">
    <svg class="logo-left" width="100px" height="100px">
      <defs></defs>
      <g id="Page-1" stroke="none">
        <g id="Switch-Logo-Red">
          <g id="Logo" transform="translate(50%, 50%) rotate(45deg)">
            <g id="Logo-Left">
              <path d="M0.2,0.2c0.2,0.2,0.2,0.2,0.2,0.2" />
              <circle id="C1" cx="0.2" cy="0.2" r="0.2" />
            </g>
          </g>
        </g>
      </g>
    </svg>
    <svg class="logo-right" width="100px" height="100px">
      <defs></defs>
      <g id="Page-1" stroke="none">
        <g id="Switch-Logo-Red">
          <g id="Logo" transform="translate(50%, 50%) rotate(45deg)">
            <g id="Logo-Right">
              <path d="M15.8,15.8c0.2,0.2,0.2,0.2,0.2,0.2" />
            </g>
          </g>
        </g>
      </g>
    </svg>
  </div>
  <div class="nintendo-switch">
    <svg width="127px" height="36px">
      <defs></defs>
      <g id="Page-1" stroke="none">
        <g id="Switch-Logo-Red">
          <g id="Logo" transform="translate(50%, 50%) rotate(45deg)">
            <g id="Logo-Left">
              <path d="M0.2,0.2c0.2,0.2,0.2,0.2,0.2,0.2" />
              <circle id="C1" cx="0.2" cy="0.2" r="0.2" />
            </g>
          </g>
        </g>
      </g>
    </svg>
  </div>
</div>

```

CSS ▼

```

html,
body {
  width: 100%;
  height: 100%;
}

.container {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  width: 100%;
  height: 100%;
}

.logo {
  margin-bottom: 1em;
}

.logo-left,
.logo-right {
  display: inline-block;
  position: relative;
}

@keyframes logo-anim-left {
  from { top: 0; }
  26% { top: 0; }
  29% { top: 7px; }
  40% { top: 0; }
}

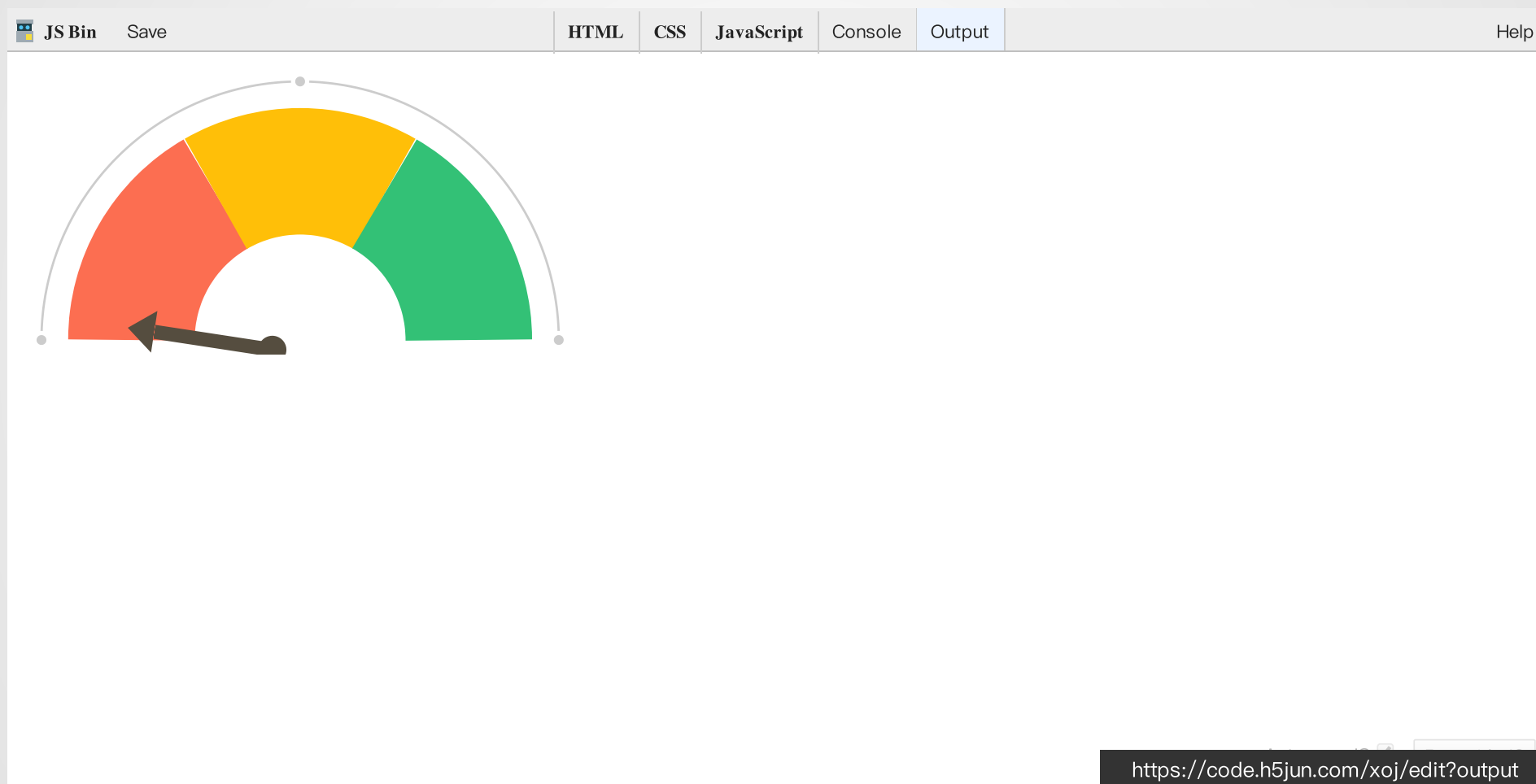
@keyframes logo-anim-right {
  from { top: -45px; }
  25% { top: -50px; }
}

```



<https://code.h5jun.com/gowis/edit?html,css,output>

旋转变换



参考阅读: [环形日期选择器](https://code.h5jun.com/xoj/edit?output)



静态资源库 (/) Google 字体库 (/index/fonts)

博客 (<https://75team.com/cate/75cdn>)

前端静态资源库

常用类库

zepto

Zepto.js is a minimalist JavaScript library for modern browsers, with a jQuery-compatible API

(/zepto)

require.js

RequireJS is a JavaScript file and module loader. It is optimized for in-browser use, but it can be used in other JavaScript environments, like Rhino and Node

<https://cdn.baomitu.com/>

☐ 隐藏非必要元素

☐ 标记旋转圆心

左前腿：

右前腿：

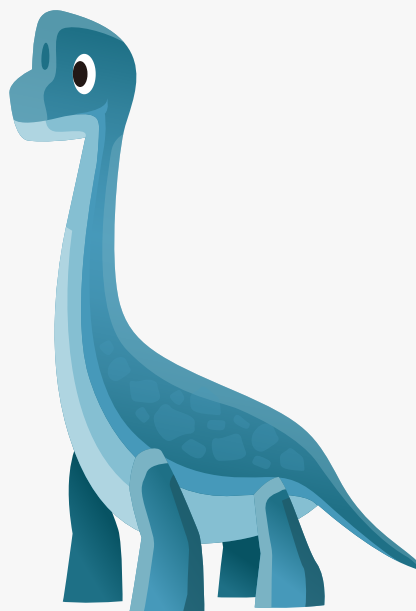
左后腿：

右后腿：

奔跑类型：

☐ 撒欢跑

☐ 小碎步



缩放变换

斜歪变换

矩阵变换

$$\begin{bmatrix} a & c & e \\ b & d & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + cy + e \\ bx + dy + f \\ 0 + 0 + 1 \end{bmatrix}$$

<https://latex.codecogs.com/svg.latex?%5Cinline%20%5Chuge...>

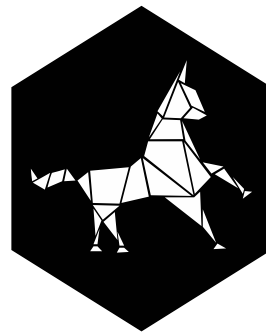
{"code":"40510004","msg":"invisible domain"}

<https://sfault-image.b0.upaiyun.com/109/394/1093948114-591...>

参考： [从矩阵与空间操作的关系理解CSS3的transform](#)



github.com/firekylin/firekylin)



Firekylin

A Simple & Fast Node.js Blogging Platform Base On ThinkJS & React & ES2015+

GET GUIDE

QUICK START

([HTTPS://GITHUB.COM/FIREKYLIN/FIREKYLIN/blob/master/DEVELOPER/LABS/LAB/10/README.md](https://github.com/firekylin/firekylin/blob/master/DEVELOPER/LABS/LAB/10/README.md))

OPEN CHAT ([HTTPS://GITTER.IM/FIREKYLIN/FIREKYLIN](https://gitter.im/firekylin/firekylin))

<https://firekylin.org>

分解动作：

- ☐ Node 背景
- ☐ 碎片环绕
- ☐ 破镜重圆

pause

变形动画(Morph)

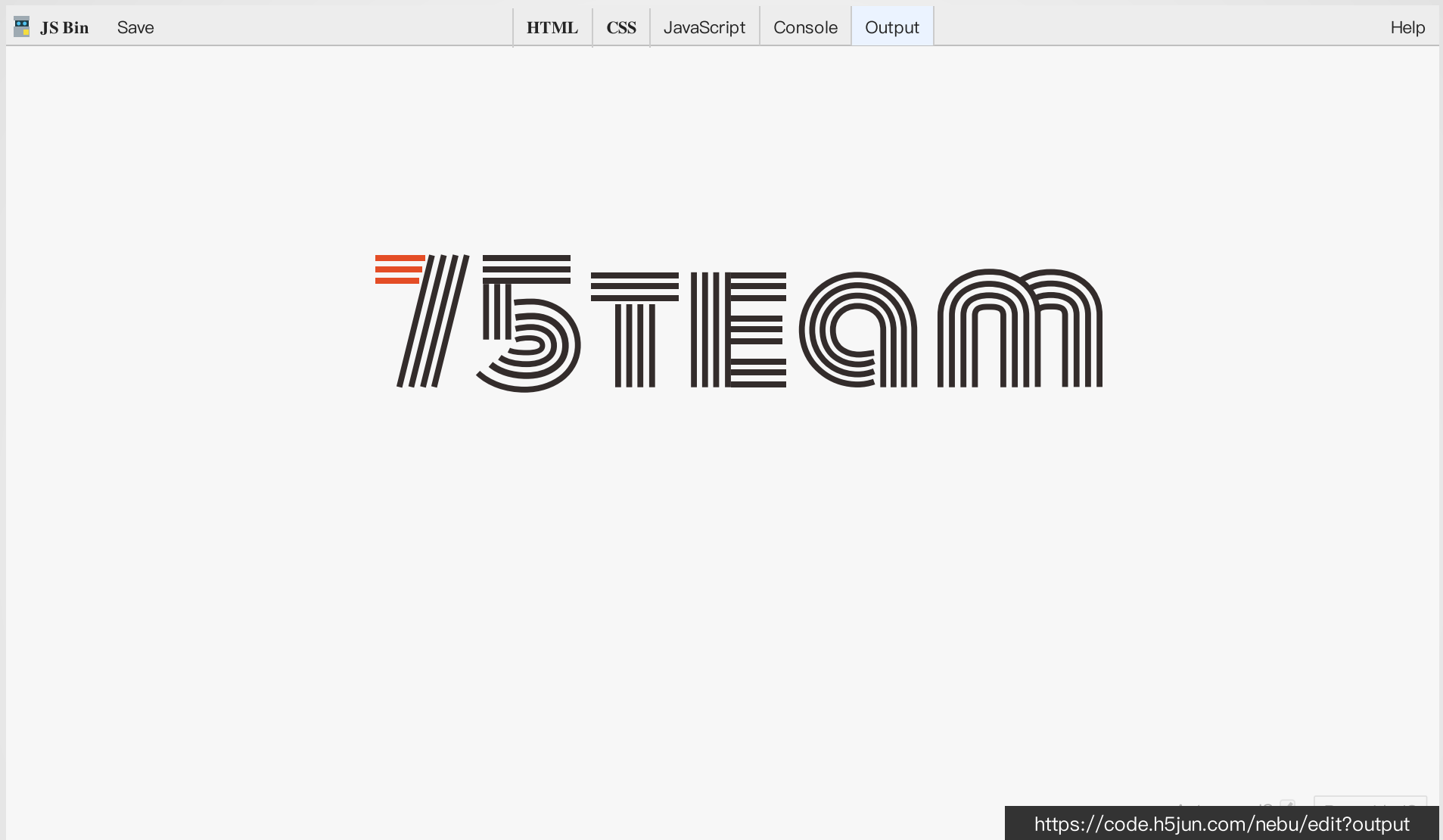
即非线性变换动画，常见类型有：

1. 描边动画
2. 路径轨迹动画
3. Path变形动画

描边动画

通过修改路径的 `stroke-dashoffset` 实现的动画

- CSS 动画
- JS [轻量 SVG 描边动画库](#)



more example: [Flower SVG](#) | [Alex Aloia LOGO](#) | [Bayleys](#)

路径轨迹动画

元素沿着某条路径运动的动画

- SVG 动画标签 `animateMotion`
- JavaScript 动态计算
- CSS `offset-path` 属性



Path变形动画

一条路径变换成另一条路径的动画

- CSS 对路径 d 属性进行动画
- [Snap.svg](#)

JS Bin

Save

HTML

CSS

JavaScript

Console

Output

Help

HTML ▼

```
<meta name="description" content="
<label>
  <input type="checkbox" id="stat
  <svg width="108px" height="108px"
    <circle id="bg-circle" stroke="
    <path class="left"/>
    <path class="right"/>
  </svg>
</label>
```

CSS ▼

```
body {
  width: 100%;
  text-align: center;
}

#status {
  display: none;
}

.left, .right {
  fill: #FFF;
  transition: all .5s;
}

.left {
  d: path('M0 1 0 49 22 38.183075
}

.right {
  d: path('M44.096035 24.8759317
}

#status:checked + svg .left {
  d: path('M0 1 0 49 10 49 10 12
}

#status:checked + svg .right {
  d: path('M35 0 35 50 25 50 25 0
}
```



<https://code.h5jun.com/mugi/edit?html,css,output>

5

- greensock
- SVG Morpheus
- Pasition

填充(Pattern), 遮罩(Mask), 滤镜(Filter)等等...
还有更多的效果等待着你们的发现!

只要有想法，不需要复杂的动效我们也能做出好看的动画。

——尼古拉斯·D·赵四

练习

给下面的 LOGO 设计动效

JS Bin

Save

HTML

CSS

JavaScript

Console

Output

Help

HTML ▼

```
<meta name="description" content="3
<?xml version="1.0" encoding="UTF-8
<svg width="200px" height="200px" v
<g id="360">
  <path d="M0.602409639,100 C0.60
  <path d="M20.9823694,39.6913982
  <path d="M181.389489,157.072822
  <g id="plus" transform="transla
    <path d="M13.288989,32.8957
    <path d="M13.288989,32.8957
  </g>
</g>
</svg>
```



<https://code.h5jun.com/malu/edit?html,output>

学习资料:

- [SVG动画实践@CSS Conf 2016](#)
- [SVG Tutorial](#)
- [张鑫旭的技术博客](#)
- [w3cplus](#)
- [SVG开发与应用](#)
- [SVG动画精髓](#)
- [@优秀网页设计](#)
- [codepen.io](#)